Practical week-3

TASK-1

Develop a C program using fork() that creates a parent and child process. Display the Process ID(PID),Parent Process ID(PPID),and process states at different stages of execution.

satwika@SATWIKA: ~/task1.c

```
  GNU nano 8.7.1                task1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main(){
pid_t pid;
printf("Parent process started\n");
printf("Parent PID :%d\n",getpid());
printf("Parent PPID :%d\n",getppid());

pid=fork();

if(pid<0){
perror("fork failed");
return 1;
}
else if(pid==0){
printf("\n--- Child Process ---\n");
        printf("Child PID   : %d\n", getpid());
        printf("Child PPID  : %d\n", getppid());
        printf("Child State: Running\n");

        sleep(5);

        printf("Child State: Terminating\n");
        exit(0);
    }

    else
    {
        // Parent process
                          [ Read 46 lines ]
^G Help          ^O Write Out     ^F Where Is
^K Cut           ^T Execute       ^X Exit
^R Read File     ^\ Replace       ^U Paste
^J Justify
```

satwika@SATWIKA: ~/task1.c

```
    else
    {
        // Parent process
        printf("\n--- Parent Process ---\n");
        printf("Parent PID  : %d\n", getpid());
        printf("Parent PPID : %d\n", getppid());
        printf("Parent State: Running\n");

        printf("Parent State: Waiting for child...\n");
        wait(NULL);
 printf("Parent State: Child terminated\n");
        printf("Parent State: Terminating\n");
    }

    return 0;
}
^G Help          ^O Write Out     ^F Where Is
^K Cut           ^T Execute       ^X Exit
^R Read File     ^\ Replace       ^U Paste
^J Justify
```

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ nano task1.c
satwika@SATWIKA:~$ gcc task1.c -o task1
satwika@SATWIKA:~$ ./task1
Parent process started
Parent PID :656
Parent PPID :356

--- Parent Process ---
Parent PID  : 656
Parent PPID : 356
Parent State: Running
Parent State: Waiting for child...
```

```
--- Child Process ---
Child PID   : 657
Child PPID  : 656
Child State: Running
Child State: Terminating
Parent State: Child terminated
Parent State: Terminating
```

TASK-2

To observe the different process states-Ready, Running, Waiting, and Terminated-using Linux monitoring tools such as ps, top, and /proc.

satwika@SATWIKA: ~/process_states.c

```
  GNU nano 8.7.1                process_states.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Process started!\n");
printf("PID: %d\n",getpid());
printf("Running state...\n");
sleep(2);

printf("Waiting/Sleeping state...\n");
sleep(10);

printf("Process terminating...\n");

return 0;
}

                                  [ Read 15 lines ]

^G Help          ^O Write Out      ^F Where Is
^K Cut           ^T Execute        ^X Exit
^R Read File     ^\ Replace        ^U Paste
^J Justify
```

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ gcc process_states.c -o process_states
satwika@SATWIKA:~$ ./process_states
Process started!
PID: 832
Running state...
Waiting/Sleeping state...
Process terminating...
```

Sol:

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ ps -o pid,ppid,state,cmd -p 832
    PID    PPID S CMD
satwika@SATWIKA:~$ top
top - 13:56:56 up 51 min,  1 user,  load average: 0.00, 0.02, 0
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem :   3758.6 total,   3113.5 free,    462.1 used,    258.
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3296.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
      1 satwika   20   0   24164  15472  11632 S   0.0   0.4
      2 satwika   20   0    3180   2200   2072 S   0.0   0.1
      7 satwika   20   0    3180   2048   1940 S   0.0   0.1
     56 satwika   19  -1   50428  16784  15484 S   0.0   0.4
     88 systemd+  20   0   22416  14460  11968 S   0.0   0.4
    103 satwika   20   0   35312  12228   9188 S   0.0   0.3
    120 satwika   20   0    2888   1948   1820 S   0.0   0.1
    121 satwika   20   0    4472   3132   2884 S   0.0   0.1
    123 message+  20   0    8820   5344   4676 S   0.0   0.1
    155 satwika   20   0   44096  29764  14624 S   0.0   0.8
    162 satwika   20   0   18424   9440   8204 S   0.0   0.2
    177 syslog    20   0  220548   5932   4648 S   0.0   0.2
    220 _chrony   20   0   23892  11340   7136 S   0.0   0.3
    236 _chrony   20   0   12096   2460   1812 S   0.0   0.1
    267 satwika   20   0    5492   2768   2584 S   0.0   0.1
    269 satwika   20   0  123460  32676  18060 S   0.0   0.8
    353 satwika   20   0    3184   1104    980 S   0.0   0.0
    354 satwika   20   0    3200   1244   1108 S   0.0   0.0
    356 satwika   20   0    6268   5528   3856 S   0.0   0.1
    357 satwika   20   0    8644   5536   4696 S   0.0   0.1
    406 satwika   20   0   22340  13272  10992 S   0.0   0.3
    408 satwika   20   0   22916   4076   2076 S   0.0   0.1
    440 satwika   20   0    6136   5416   3864 S   0.0   0.1
    902 satwika   20   0    8164   5996   3896 R   0.0   0.2
```

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ ./process_states &
[3] 1261
Process started!
PID: 1261
Running state...
satwika@SATWIKA:~$ Process terminating...
Waiting/Sleeping state...
ps -o pid,ppid,state,cmd -p 1261
    PID    PPID S CMD
   1261     356 S ./process_states
[2]-  Done                    ./process_states
```